

Rajiv Patil

Public-Space Crowd Supervisor

About

Age: 42
Location: Railway / metro station
Occupation: Crowd Supervisor
Education: Not specified

Coordinates crowd-management activities in platforms, entrances and waiting zones.

Monitors busy zones, coordinates teams and responds to changing crowd conditions.

Personality & preferences

Situational focus | Live heatmaps | Clear warnings | Fast zone access | Reliable updates | Privacy-first

Motivations

Better situational awareness and smoother public movement. Rajiv needs a live floor-plan heatmap, clear high-density warnings, fast zone selection and reliable updates.

Goals

- Detect rising crowd density early.
- Coordinate crowd-management measures.
- Coordinate operational teams.

Pain points

- Crowds change quickly.
- Manual observation covers limited areas.
- Individual tracking may be intrusive.

Illustrative user quote

"Show me which zones need attention right now, without tracking individual passengers."

Expected System Features

- Live station heatmap with quick zone selection.
- ESP32 zone-node online/offline indicators.
- Raspberry Pi5 local-processor/data-feed availability status.
- Connection-loss and stale-data warnings.

Success Criteria

- Distinguish low density from missing data.
- Identify affected zones and escalate faults promptly.
- Resume confident zone monitoring after technical staff restore updates.

Personality

Operational focus	Flexible / Reactive
Alert-driven	Proactive planner
Privacy sensitive	Data open
Tech cautious	Tech savvy
Slow coordinator	Fast coordinator

Technical skills

Heatmap reading	●●●○○
Zone selection	●●●○○
Alert interpretation	●●○○○

Tools or platforms

- Floor-plan heatmap
- Density warnings
- Zone overview

Practical-1



Amit Deshmukh

College Facility Manager

About

Age: 38
Location: College campus
Occupation: Facility Manager
Education: Not specified

Responsible for classrooms, corridors, libraries and canteen areas.

Checks facility reports and coordinates campus staff.

Personality & preferences

Operational focus
Simple heatmaps
Clear density labels
Alert-driven
Quick zone access
Privacy-first

Motivations

Safer, smoother campus operations and less manual monitoring. Simple heatmaps, clear density categories, alerts and quick zone access help Amit coordinate staff.

Goals

- Identify crowded zones early.
- Coordinate staff efficiently.
- Improve campus movement.

Pain points

- Manual headcounts take time.
- Crowding can be noticed late.
- Camera-based monitoring raises privacy concerns.

Illustrative user quote

"I need to know which campus areas are getting crowded before the situation becomes difficult to manage."

Expected System Features

- Live campus zone heatmap.
- Low/medium/high estimated-density labels.
- Simple high-density alerts.
- Last-update time and unavailable-zone indicator.

Success Criteria

- Locate zones needing attention quickly.
- Coordinate campus staff with clear zone information.
- Reduce dependence on routine manual checks.

Personality

Operational focus	Flexible / Reactive
Alert-driven	Proactive planner
Privacy sensitive	Data open
Tech cautious	Tech savvy
Slow coordinator	Fast coordinator

Technical skills

Dashboard reading ●●●○○
Heatmap interpretation ●●●○○
Staff coordination tools ●●○○○

Tools or platforms

- Computer dashboard
- Zone heatmap
- Facility reports



Neha Kulkarni

Hospital Operations Coordinator

About

Age: 34
Location: Hospital
Occupation: Operations Coordinator
Education: Not specified

Coordinates operations in waiting areas and shared public zones.

Reviews operational information, coordinates staff and checks changing conditions.

Personality & preferences

Operations focus
Readable heatmaps
Aggregate estimates
Historical trends
Local processing
Privacy-first

Motivations

Better operational awareness and privacy-conscious monitoring. Neha needs a readable zone heatmap, anonymous aggregate estimates, historical trends and local processing.

Goals

- Understand crowding in waiting areas.
- Coordinate available resources.
- Reduce avoidable congestion.

Pain points

- Waiting areas crowd unexpectedly.
- Manual observation is inconsistent.
- Patient privacy must be protected.

Illustrative user quote

"Help me understand where crowding is increasing without exposing anyone's identity."

Expected System Features

- Waiting-area zone shortcuts.
- High-density alerts with clear zone labels.
- Estimated historical density trends.
- Visible refresh/stale-data status.

Success Criteria

- Identify which waiting areas need staff attention.
- Understand crowd categories at a glance.
- Review changes without exposing patient details.

Personality

Operational focus	Flexible / Reactive
Alert-driven	Proactive planner
Privacy sensitive	Data open
Tech cautious	Tech savvy
Slow coordinator	Fast coordinator

Technical skills

Operations reports ●●●○○
Heatmap reading ●●●○○
Trend interpretation ●●○○○

Tools or platforms

- Zone heatmap
- Historical trends
- Operations reports